

Abduct, Deduct, Induct, Respawn!

A Retroductive Perspective on Games

Anne Vetter, Sebastian von Mammen
Games Engineering, University of Würzburg, Germany
{anne.vetter, sebastian.von.mammen}@uni-wuerzburg.de

Abstract—Games are more than systems to master; they are engines of iterative thought. Through play, users engage in a retroductive loop: an unexpected event triggers hypothesis formation (abduction), leading to action planning and execution (deduction) and learning from feedback (induction). While it is central to player experience, current game AI typically optimizes for challenge balancing or content generation, rarely reflecting back on the player’s evolving mental model. We propose treating both the player and the game as coupled retroductive agents that continuously observe, hypothesize, act, and learn across nested timescales. We formalize the retroductive processes of each agent and model their co-evolutionary dialogue as an exchange of inferences and feedback, positioning the game not as a reactive system but as a thinking partner. Building on this formalism, we articulate the vision of retroductive play: Gameplay as an epistemic negotiation in which the system deliberately probes, challenges, and reshapes the player’s mental model while refining its own, judging progress by shifts in understanding rather than optimized performance. Retroductive play invites players—and the game that serves them—to respawn their ways of thinking.

Index Terms—Game Design, Player Modeling, Artificial Intelligence in Games, Human–Computer Interaction, Gameplay Mechanics, Reasoning

I. INTRODUCTION

Before you master Mario, you think like Peirce. Playing video games is a form of active reasoning, mirroring the iterative cycle—abduction, deduction, and induction—that Peirce [1] called “retroduction”, and which underpins how players make sense of games. While research has explored game-based learning (e.g., [2]–[4]), engagement (e.g., [5]), and adaptive game systems (e.g. [6], [7]) these areas are often treated in isolation. A unified cognitive account of how players make sense of games and how systems might mirror this reasoning to enable reciprocal dialogue—remains underexplored.

This paper aims to address this gap by proposing a dual retroductive perspective, positioning both player and game as reasoning agents. Our contributions are as follows: (1) We define the player’s retroductive cycle as the core cognitive mechanism driving gameplay. (2) Central to our thesis, we ‘flip the perspective’ to conceptualize the game as a retroductive reasoner, detailing its core components and reasoning loop. (3) We model the coupled interaction between both, introducing a set of factors that shape the emergent co-adaptive dynamics. (4) We derive a set of design provocations aimed at fostering “retroductive play,” where the game challenges the player to think differently.

By reframing player–game interaction as a mutual reasoning process, we present the player and the game as adaptive reasoning partners, each modeling and responding to the other across the interface. This perspective bridges game design, cognitive science, and AI, positioning games as engines of iterative thought for both humans and intelligent systems.

Section II outlines the foundations of human reasoning and retroduction. Section III presents the player’s retroductive loop. Section IV explores the game as a retroductive agent. Section V formalizes the coupled dynamics between both agents. Section VI proposes design provocations for retroductive play. Finally, we conclude with a summary and directions for future research.

II. THE FUNDAMENTALS OF HUMAN REASONING

Gameplay is a cognitive process of iterative sense-making and problem-solving. This section outlines its theoretical foundations in human reasoning.

Reasoning is a complex cognitive process involving the integration of existing knowledge with new information to generate insights and conclusions [8], [9]. Neuroscientifically, this capacity is supported by the fronto-parietal Multiple Demand (MD) system. Core MD regions—such as the dorsolateral prefrontal cortex, anterior cingulo-insular areas, and intraparietal sulcus—enable flexible cognition by prioritizing task-relevant inputs, updating working memory, and configuring connectivity for novel problem demands [8]. This dynamic control underlies the mental modeling and restructuring central to mental model theory and retroduction, discussed below. Among various psychological theories of reasoning, mental model theory provides a robust and empirically supported framework [9], [10]. This theory posits that humans reason not by applying formal logical rules, but by constructing and manipulating internal representations, mental models, of situations, possibilities, or systems [11]. These models are dynamic, allowing for defeasibility (the revision of conclusions when new, contradictory information emerges) and non-monotonicity (where adding new information can alter, rather than merely extend, existing conclusions), reflecting the flexibility of real-world reasoning [12]. This is particularly relevant to spatial reasoning, which involves the construction and online navigation of relational spatial models [13], a skill that interactive environments can engage and potentially enhance [14]. Mental model theory also aligns with dual-process theories of cognition [12], which distinguish between two

systems of thought. System 1 operates quickly and intuitively, often relying on heuristics and processing a single mental model at a time due to limited access to working memory. In contrast, System 2 is slower, more deliberate, analytical, and capable of constructing and evaluating multiple mental models by engaging working memory, thus supporting more complex and novel problem-solving [15]. These qualities make mental model theory a strong account of how humans navigate uncertainty and complexity in reasoning tasks [12].

Reasoning, per Peirce [1], involves three interconnected modes: **Deduction** ($Rule + Case \rightarrow Result$) applies general rules to a specific case yields a necessary result; **Induction** ($Case + Result \rightarrow Rule$) generalizes rules by observing repeated cases and their results; and **Abduction** ($Rule + Result \rightarrow Case$), where when confronted with a surprising result, we infer a plausible case—a hypothesis—that, under existing rules, would explain it [16]. Unlike the others, abduction introduces new ideas [1], [16], [17], making it central to creative thought and the evolution of mental models. Magnani further distinguishes a purely theoretical abduction, an inner, model-based search, from manipulative abduction, where “thinking-through-doing” enlists bodily action and external artifacts to elicit information contemplation could never supply. Because these external artifacts act as epistemic mediators, the mental models we reason with are not confined to the head; they sprawl across icons, images, bodily movements, and tools, dynamically redistributing cognitive load [18], [19]. The reasoning modes are not isolated functions but are dynamically intertwined in a continuous, self-correcting, cycling meta-reasoning process termed “retroduction” by Peirce [1]. A surprising observation triggers abduction, which proposes a hypothesis, a tentative adjustment to our mental model. This hypothesis is tested deductively by deriving implications and predictions, which are then evaluated inductively. The results inform whether the hypothesis is confirmed, revised, or rejected, thereby refining the underlying model [1], [16]. Retroduction reflects scientific reasoning: the iterative cycle of observation, hypothesis formation, experimentation, and theory revision that drives the evolution of scientific knowledge [20], [21]. In obvious cases, retroduction’s steps may unfold so rapidly, that they appear as a singular and instantaneous conscious judgment [20].

III. RETRODUCTION AS THE NATIVE GRAMMAR OF PLAY

Our innate ability for retroductive reasoning, discussed in Section II, finds particularly fertile ground for expression and exercise within the interactive environments of video games. Play, at heart, is a continuous act of making sense out of novel situations, driven by this cycle of hypothesizing, testing, and learning. While human players engage in these cycles intuitively, unpacking how games prompt and sustain them is essential for both appreciating their cognitive richness and crafting compelling interactive experiences. This section aims to formalize the players’ interaction with the game as a retroductive process, employing a mapping of game components and gameplay loops to the reasoning concepts.

A. Defining the Game

To formalize the player–game interaction, we adapt a model of games presented by Francillette et al. [22], who draw from Chauvier’s work [23]. Chauvier’s model defines a game through core components including actions, states, rules, goals, players, and challenges [22], [23]. Building on this foundation, we introduce an additional component, context channels, inspired by Schell’s taxonomy of game elements [24]. Therefore, we conceptualize a Game (G) as a structured interactive system, represented as a tuple:

$$G = \langle S, P, A, R, \Gamma, C, \Xi \rangle$$

The elements states S , players P , actions A , rules R , goals Γ , challenges C derive from the original formulation. Our extension, Ξ , represents the multifaceted context channels.

A game progresses through a series of **states** (S), where each state ($s \subseteq S$) represents a snapshot of all relevant game variables at a given moment. The initial state s_0 , serves as the initial *result* or observation that prompts the player’s first abductive inferences. The game world is populated by **players** P , embodied, intentional agents (human or artificial) who possess internal mental models M that encapsulate their current understanding of its rules and dynamics and reasoning capacities. While P can include multiple co-existing players, our analysis of the player’s engagement isolates the retroductive reasoning cycle of the primary interacting agent at a given moment, termed the Player-Agent (GP), who possesses their mental model (M^P). Players interact with the game through a set of available in-game **actions** A . An action $a \in A$ transitions the game from one state s to another s' . With this primary interface between the player and the game system, hypotheses are tested, while outside-game theoretical actions such as strategic planning [25] can prune or reframe the perceived action space. The game’s behavior is governed by a set of **rules** R , which are formal constraints and affordances defining valid state transitions ($s' = R_{game}(s, a)$) and the underlying logic of the game world. These rules are often initially unknown or only partially known to the player and become the target of their abductive and inductive reasoning. Understanding these rules forms the basis of the player’s mental model. Player activity is typically directed towards achieving specific **goals** $\Gamma \subseteq S$, which are desirable states. Goals provide direction and motivation for the player’s retroductive process, framing the desired results that guides their hypothesis formation and action selection. These goals can be explicit and system-defined or emergent and player-defined, especially in less constrained games. **Challenges** C are obstacles, uncertainties, or complexities within the game system that impede the player’s progress toward their goals. Challenges are often the “surprising facts” that trigger abductive reasoning and necessitate problem-solving through retroductive cycles. While mechanical aspects are already embedded in the prior components, we introduce Ξ as **Context channels** to account for the multifaceted presentation layers through which the game world is experienced. The aesthetic dimension shapes

the player's sensory immersion through visuals, audio, and style; its narrative weaves thematic and plot elements that give meaning to play; and its technology delivers the entire package through software and hardware, which enables and constrains the entire interactive system [24].

B. The Player's Retroductive Gameplay Loop

Gameplay structure has been modeled through many frameworks like Mechanics–Dynamics–Aesthetics (MDA) [26], which offers a designer-focused lens, or Objective–Challenge–Reward (OCR) [25]. The OCR model adeptly explains why players might be motivated by objectives, challenges, and rewards, but is less explicit about how they cognitively process novel situations, formulate strategies to overcome challenges, or adapt their understanding from feedback, particularly in the rapid, iterative micro-interactions where discrete rewards may be absent or subtle. As Guardiola [25] suggests, this model can be too “macro” to capture the nuanced activity of gameplay.

A more cognitively grounded *Retroductive Gameplay Loop* can be formalized and illustrated through a familiar example from *Minecraft* [27], a hugely successful block-based open-world sandbox and adventure game, as follows:

Observation of Surprising Result (Abductive Trigger):

In state s_t , the *GP* mines an Iron-Ore block with a wooden pickaxe and observes an unexpected *result*

$$Res_t = \text{“no iron drops”}.$$

The event clashes with the *GP*'s current *mental model* M_t^P (“every block yields itself when broken”), signaling a knowledge gap that launches a retroductive cycle.

Hypothesis Generation (Abduction): To resolve this surprise, the player proposes a hypothesis H_t that would explain Res_t under the still-trusted rules:

$$(M_t^P \wedge H_t) \vdash Res_t$$

where $H_t = \text{“wooden tools are too weak to harvest iron.”}$

Abduction inserts H_t into the mental model as a tentative *case*.

Action Planning & Execution (Deduction): To test H_t , the *GP* plans (which is an outside-game action) one or a series of in-game actions ($a_t \in A$). This deductive step is guided not only by H_t but crucially by their pre-existing M_t^P which includes an implicit understanding of the game's tiered material progression (Wood $\rightarrow$ Stone). The most logical action is therefore to craft a stone pickaxe and mine again. Deductively, the *GP* predicts that this action will yield a different result:

$$(M_t^P \wedge H_t \wedge a_t) \vdash Res_{pred}, \quad Res_{pred} = \text{“iron drops.”}$$

Executing a_t transitions the game to s_{t+1} .

Outcome Evaluation & Learning (Induction) In s_{t+1} , *GP* observes the outcome (Res_{t+1}) of a_t and compares Res_{t+1} with Res_{pred} . Induction updates the rule set:

$$(a_t \wedge Res_{t+1}) \vdash M_{t+1}^P.$$

If $Res_{t+1} = Res_{pred}$ (iron drops), H_t is confirmed and the new rule “tool tier $\geq$ stone required for iron” is integrated; otherwise H_t is weakened or discarded, and the cycle restarts.

The retroduction cycle repeats iteratively, refining the M^P and approach through a spectrum of nested loops that unfold across varying timescales. Beatty's multi-scale model of gameplay [28] provides the granularity we need for mapping retroduction onto nested loops in a game's structure. In his framework, every game unfolds through four concentric “games”—micro, macro, meta (builder, and social)—each driven by its feedback loops of exploration, motivation, and identity development. We borrow this taxonomy of scales, extend it by reading each loop as a retroductive cycle, and supply concrete gaming examples.

At the most immediate level, *micro loops* capture rapid, intuitive interactions guided predominantly by System 1 processing. These loops occur at the immediate, moment-to-moment level – timing a jump precisely in *Celeste* [29], rotating a *Tetris* [30] piece, or making quick tactical adjustments in *StarCraft* [31]. Such loops rely heavily on practiced heuristics and spatial intuition, allowing *GPs* to rapidly hypothesize, act, and revise based on immediate feedback. Through repeated exposure, these micro-loops refine motor skills, perceptual acuity, and intuitive understanding of game mechanics. Each such loop refines a local fragment of the mental model M_t^P . Expanding in scale, *macro loops* involve prolonged, deliberative, and strategic reasoning, engaging System 2 processes. These loops encompass higher-order gameplay goals and strategies, such as resource management over multiple turns in *Civilization* [32], puzzle solving across interconnected stages in *The Legend of Zelda* [33], or planning and executing complex builds in *Minecraft* [27]. Macro loops synthesize the pattern knowledge acquired in micro loops into coherent, high-level mental models and stepwise plans. Beyond direct gameplay, *meta-level loops* encompass both game modification and social interaction. In the *Build-the-Game loop*, *GPs* creatively abduct and test new mechanics or rules by modifying the game itself – designing redstone circuits, custom maps, or balance patches [34]. These innovations often seed the *Play-the-Culture loop*, where larger communities engage in the generation of shared hypotheses, the crafting of theories, and the refinement of collective models via forums, streaming, and competitive scenes [34]. This highlights how reasoning extends beyond individual cognition, evolving into a distributed, cultural process of meaning-making and co-creation. This distributed sense-making is not without risks, however. The same powerful, recursive patterns that allow for collective creation are also the mechanisms that can produce online echo chambers, where polarized groups reinforce their own beliefs and grow resistant to outside perspectives [35]. This creates a need for communities to develop cognitive strategies to guard against such effects. One such strategy is a form of psychological inoculation, where preemptively exposing individuals to a weakened form of a misleading argument or social dynamic helps build mental resistance against future manipulation [36].

Thus, reasoning cascades both ways: the upper layers set targets and framing assumptions for the lower layers, and the lower layers supply the empirical substrate that justifies revisions up the chain. Therefore, micro-level experiences directly inform higher-order macro hypotheses; strategic macro objectives guide immediate micro decisions. Mastery at the macro level often inspires players to innovate within the game structure, driving game-modification activities. Successful modifications can influence community practices, evolving into broader social interaction loops.

The precise manifestation and complexity of these retroductive loops depend significantly on the specific constraints of each game. Highly structured games like *Chess* or *Tetris* [30] foster tightly focused reasoning loops, driven by clear rules and immediate, deterministic feedback. Conversely, open-ended environments such as *Minecraft* [27] encourage exploratory loops, with numerous low-level experiments aggregating into extensive strategic insights and creative discoveries.

Thus, retroduction is foundational to play. Games are environments that inherently engage players in this cycle of inquiry, providing a “curriculum” for critical thinking and problem-solving. This active learning allows players to construct meaning and develop “epistemic frames” that may transcend the game itself [3].

IV. FLIPPING THE PERSPECTIVE: THE GAME AS A RETRODUCTIVE AGENT

Just as the *GP* reasons about the game in recurring retroductive cycles, a modern video game increasingly reasons about its players. In this section, we conceptualize the game itself as a retroductive agent. We begin by defining the core components of such a “Game-as-an-Agent” (*GA*), focusing on its perceptual capabilities, its internal model, and its capacity to act upon itself. We then detail how these components enable the *GA* to engage in its own retroductive loop, mirroring the *GP*’s cognitive processes to achieve adaptive gameplay.

A. The Game-as-an-Agent Components

To capture this symmetry between *GP* and *GA*, we further extend the game definition $G = \langle S, P, A, R, \Gamma, C, \Xi \rangle$ to a fit a retroductive *Game-as-an-Agent* as follows:

$$GA = \langle S, P, A, R, \Gamma, C, \Xi, M_G, \mathcal{O}, \mathcal{U} \rangle.$$

We introduce the game’s own perception module ($\mathcal{O}$), “mental model” (M_G), and action space ($\mathcal{U}$), each detailed below.

The *GA*’s **perception module** ($\mathcal{O}$) acts as its senses, formally a mapping $\mathcal{O}: P \times S \rightarrow \Sigma$ that for a given *GP* $p \subseteq P$ and game state s converts multimodal streams of telemetry (e.g., inputs, timings, deaths, biometric proxies, social signals) [37], [38] into structured observations Σ , compact latent representations [39] suitable for inference. For example, multiple “player death” events at one location could become a “high death frequency at location X” observation. Modern systems leverage cloud infrastructure and big data analytics for processing vast data volumes generated by players [40].

In multiplayer settings, this capability is crucial for handling complex interaction [41].

These observations feed into the *GA*’s “**mental model**” (M^G), a mutable latent belief state of player skill, preference, and the system’s own efficacy. Active player modeling, where systems strategically learn about users, is an established concept [42], [43]. Williams et al. [44] advocate for a comprehensive player profile framework that synthesizes demographic as well as psychographic data (considering personality and preferences), mental models, i.e. the player’s understanding of game mechanics and genre, and behavioral models (derived from game analytics). Their work suggests that integrating these diverse data streams can lead to more robust and predictive player profiles, e.g. combining self-efficacy measures with game performance data can better explain player behavior. For multiplayer online games, M^G should be extended beyond individual player characteristics – such as skill levels, behavioral patterns like aggression or exploration, and strategic intentions – to include team dynamics and population-level trends [45]. Further, emerging AI awareness concepts, as reviewed by Li et al. [46], suggest future M^G components like meta-cognition, the ability to represent and reason about its own states, such as its confidence in a prediction of the *GP* behavior or the uncertainty of its current M^G . This could lead to *GAs* being aware of the limitations of their M^G , leading to more cautious or exploratory adaptive interventions, particularly in complex or novel player interaction scenarios.

Based on $\mathcal{O}$ and its current M^G , *GA* can choose to act. Its **action space** $\mathcal{U}$ is a set of adaptive interventions, functions that can modify any subset of the core game components of G :

$$\mathcal{U} \subseteq \{ u : \langle S, P, A, R, \Gamma, C, \Xi \rangle \longrightarrow \langle S, P, A, R, \Gamma, C, \Xi \rangle \}.$$

These interventions allow the *GA* to dynamically alter its own structure and behavior. As examples, consider the following functions which aim at each of the target sets individually: u_S modifies game states S , e.g. by means of procedural level reshaping as seen in *No Man’s Sky* [47] or adapting level features for target difficulty [7]. The composition of players P within a session is managed by u_P interventions, such as *Overwatch*’s [48] role-queue system. Research in this area enhances matchmaking beyond simple skill ratings, for example, by optimizing team and position assignments using player preferences and predictive models to improve fairness and satisfaction [49]. Interventions u_R target the rules R , e.g. through balance patches, common in titles like *Street Fighter 6* [50]; Pfau and Seif El-Nasr analyze such data-driven balancing where game mechanics are adjusted based on performance data and player feedback [51]. Player goals Γ can be dynamically injected or altered using u_Γ , for example, with procedurally generated quests from simulated character desires and world states [52]. Challenges C can be tuned by u_C interventions for difficulty scaling, seen in *Resident Evil 4 (Remake)* adjusting enemy parameters or in *Left 4 Dead*’s [53] AI Director; research explores similar systems of dynamic difficulty adjustment (DDA) that modify game

parameters to match player skill or completion times [7], [54]. Finally, u_{Ξ} interventions modify context channels Ξ , such as providing color-blind palettes or enhanced subtitles to improve accessibility [55].

B. The Game's Retroductive Loop

The *GA* cycles through the same reasoning stages as the player, but with different evidence and levers. It observes the player's interaction traces, abduces possible explanations for surprising outcomes, plans corrective or exploratory interventions, and learns from the observed effects. To illustrate the full cycle from the *GA* side, consider a 2D platformer that tracks player performance on a difficult spike pit sequence:

Observation of Surprising Result (Abductive Trigger):

The perception module records $\Sigma_t = \mathcal{O}(p_i, s_t)$ and detects an unexpected pattern: the *GP* has died in the same spike pit five times within thirty seconds, a frequency far exceeding the cohort norm or system-level expectation given prior runs and calibrated difficulty.

Hypothesis Generation (Abduction): The *GA* forms a local hypothesis to account for the deviation under its current model:

$$(M_t^G \wedge H_t^G) \vdash \Sigma_t,$$

$$H_t^G = \text{"Jump-timing skill below required threshold."}$$

This abductive explanation links the observed failure to a latent parameter in the M^P —imprecise or mistimed jumps under current conditions.

Action Planning & Execution (Deduction): Under H_t^G the *GA* selects an intervention u_t that modifies the spike pit geometry – e.g., enlarging the landing platform by 20% or slightly reducing gravity at the apex of the jump zone – and predicts that this will reduce error frequency:

$$H_t^G \wedge u_t \vdash \Sigma_{\text{pred}} = \text{"< 1 death in next 30 s."}$$

The level geometry is altered in real time and the state advances to s_{t+1} . Methodologies analyzing character movement parameters (like jump height) to procedurally generate solvable levels, as explored by Brandner et al. [56] for jump'n'run games, demonstrate how environments can be systematically related to character abilities.

Outcome Evaluation & Learning (Induction): New telemetry Σ_{t+1} shows the *GP* clears the pit at the very next attempt and proceeds without further deaths. Because $\Sigma_{t+1} \approx \Sigma_{\text{pred}}$,

$$(u_t \wedge \Sigma_{t+1}) \vdash M_{t+1}^G,$$

the hypothesis H_t^G is strengthened, and M^G is updated. The adaptation might persist until sustained success prompts a re-evaluation of difficulty, potentially leading the *GA* to form new hypotheses about the *GP*'s (now improved) skill. This fosters a co-evolution where temporary scaffolds support player growth, and learned competence reshapes future challenges. Conversely, when an intervention fails to produce the predicted result, the *GA* must consider reasons beyond a simple misjudgment of *GP*'s skill. As active agents, *GPs* might

boycott assistance, or simply fail to notice or understand the change. A sophisticated *GA* treats these meta-game reactions as valuable data, forming new hypotheses about the *GP*'s deeper intentions and awareness.

Thus, the game's retroductive loop enables a structured dialogue, facilitating not just accessibility but a dynamic interplay where challenges and capabilities evolve together. Mirroring the *GP*'s own multi-scale reasoning, the *GA*-sided retroduction also operates across different scales.

At the micro scale, analogous to a *GP*'s immediate mechanical execution and reaction, the *GA* makes highly responsive adjustments. Its focus is on real-time system tweaks. These can include interventions modifying the properties or outcomes of *GP* actions (u_A), such as through adaptive aim-assist mechanisms that alter the efficacy of *GP* input. The *GA* also dynamically modifies local game states (u_S) or tunes immediate challenge parameters (u_C), for instance by causing momentary shifts in enemy AI aggression, all in direct response to the *GP*'s observed actions and performance within a specific, ongoing interaction.

On the macro scale, mirroring a *GP*'s strategic planning within a single play session or distinct game segment (like a level or match), the *GA* manages the broader experiential arc. This loop influences aspects like overall pacing or resource availability by adjusting larger game states (u_S) and challenge structures (u_C). It can also extend to dynamically altering player goals (u_G) or, in multiplayer contexts, managing the composition of players (u_P) for a session. The AI Director in *Left 4 Dead* exemplifies this by adjusting spawn waves and item placement (u_C, u_S) to shape overall tension. Similarly, *No Man's Sky* might reseed planetary missions (u_C, u_G) based on engagement patterns.

At the meta scale, analogous to a *GP*'s long-term skill development, community engagement, or overarching achievement goals, the *GA* (currently through development and live-ops teams) focuses on the sustained health and evolution of the entire game ecosystem and its player base. This loop drives systemic, longer-horizon changes by adjusting core game rules (u_R) or modifying context channels (u_{Ξ}) that affect the entire player base. Other meta-interventions involve evolving matchmaking systems (which can be systemic applications of (u_P) informed by new (u_R) or introducing major accessibility overhauls (u_{Ξ}).

In this cascaded system, micro-observations (e.g., frequent player deaths in a specific encounter) can feed into macro-hypotheses (e.g., "this player segment struggles with this type of challenge"). Successful macro-adjustments (e.g., a system that offers alternative strategies or slightly weakens a challenging encounter for a given segment) can then inform long-horizon live-service decisions or meta-level updates (e.g., a global re-tuning of that encounter, or the addition of a tutorial for that mechanic). Conversely, meta-level data, such as widespread community sentiment expressing frustration about a particular game system, can set new targets or constraints for macro-level adaptive policies. This multi-scale reasoning mirrors the *GP*'s own cognitive processes, where immediate

tactical decisions are informed by broader strategic goals, and long-term learning reframes these strategies.

Modeled this way, a *GA* becomes a co-reasoning partner: it observes, hypothesizes, acts, and learns much like its *GPs*, but on different data streams and horizons.

V. COUPLING THE TWO REASONING ENGINES

Having modeled both the player (*GP*) and the game (*GA*) as individual retroductive agents, we now turn to their interaction. While gameplay emerges from any player-game interaction, the coupling of two retroductive agents facilitates a unique dynamic: a co-evolutionary dialogue where the *GA* does not just adapt to the *GP*, but challenges them to think differently. This section formalizes this coupled relationship, showing how it gives rise to a spectrum of co-adaptive dynamics that can be designed to foster “retroductive play”—an epistemic negotiation where *GP* and system co-author an insightful experience.

The interaction, $I_{GP \leftrightarrow GA}$ between *GA*’s reasoning loop and *GP*’s loop, generates the observable dynamics of this dialogue. Drawing an analogy to user guidance frameworks where the interaction between a user and a system maps to a set of possible guidance mechanisms [57], we state that this interaction yields a comprehensive set of coupled retroductive dynamics, denoted as K :

$$I: G_P \times G_A \xrightarrow{\text{time}} K,$$

K captures evolving interaction patterns by tracing system states over time. Each point in this trajectory, $k_t = (s_t^{GP}, s_t^{GA}, t)$, at a specific time or event index t , records the concurrent states of player (s_t^{GP}) and game (s_t^{GA}), such as their mental models, goals, or game world status. The elements of K are thus classifications of these state trajectories, revealing how the *GP*’s hypotheses and actions influence the *GA*’s model and interventions, and vice versa, as their interaction unfolds. The interaction is facilitated by two iterative streams. The evidence stream $\Phi = \langle \phi_t \rangle_{t \in \mathbb{N}}$, where each $\phi_t = (\Sigma_t^P, \Sigma_t^G)$ represents a state of mutual perception at time t : what the *GP* observes of the game and what the *GA* observes about the player. The intervention stream $\Lambda = \langle \lambda_t \rangle_{t \in \mathbb{N}}$, where each $\lambda_t = (a_t^P, u_t^G)$ denotes a composite set of actions occurring at time t : the *GP*’s action on the game a_t^P and the *GA*’s adaptive intervention in return, u_t^G .

As a state transition only requires an action from one agent, and agents can operate at vastly different granularities, the two agents engage in asynchronous, event-driven loops. An action from one agent simply creates a new event and observation for the other, which may or may not trigger a full reasoning cycle. The *GP*’s loop is the primary driver of state change, with each action a_t^P enriching the evidence stream. The *GA* monitors this stream for salient patterns and, when triggered, intervenes with u_t^G to alter the game system, shaping the *GP*’s next reasoning cycle.

The characteristics of this asynchronous dialogue, and thus the types of dynamics observed within K , are shaped by several foundational coefficients. These coefficients— Latency

L , Bandwidth B , and Authority A — are designable properties or configurable parameters of the coupling mechanism (the interface and interaction protocols) between *GP* and *GA*. Any adaptive game exhibits a particular configuration of these coefficients. Latency Dynamics ($K_L \subseteq K$) encompass patterns of interaction characterized by the Latency coefficient (L). L represents the time lag (measured in terms of discrete event steps t from the streams Φ and Λ , or potentially real time) in perception and response on either side, thereby governing the tempo of the dialogue. Bandwidth Dynamics ($K_B \subseteq K$) encompass patterns characterized by the Bandwidth coefficient (B), which represents the amount or nature of reasoning content made mutually visible through Φ , thereby defining the clarity and depth of the dialogue. And Authority Dynamics ($K_A \subseteq K$) encompass patterns characterized by the Authority coefficient (A) representing the relative influence of each agent’s interventions from Λ when intentions conflict, thereby setting the power balance of the dialogue.

These three coefficients are present in any adaptive dialogue. Fully considering the retroduction model of reasoning, we introduce a fourth coefficient that defines the focal point of the adaptation itself. We refer to the resulting interaction patterns as Epistemic Target Dynamics $K_T \subseteq K$, which are characterized by the Epistemic Target coefficient (T). T specifies the particular layer of the *GP*’s experience, ranging from their observable behaviors and skill progression to their underlying cognitive states like current understanding and hypothesis, that *GA*’s interventions (observed in λ_t) are intended to influence (T). This T is not necessarily static; a sophisticated *GA* might shift its T dynamically based on its model of the *GP*’s state, e.g., shifting from a retroductive to an assistive T when high frustration is detected. The T exists on a spectrum of depth: at its most basic level, T is the *GP*’s immediate Task/Outcome, where it intervenes simply to ensure task completion, aligning with traditional user guidance. A more sophisticated *GA* might target the *GP*’s Performance/Skill, intervening to manage challenge and foster mastery, which is the focus of classic DDA systems. At the deepest level, T becomes the *GP*’s Hypothesis/Model. Here, interventions are explicitly designed to engage with the reasoning process itself by introducing surprising counter-evidence or providing scaffolding to provoke conceptual change in the M^P .

Retroductive play emerges from the interaction between a *GP* and a *GA* that is capable of and actively sets its Epistemic Target to the *GP*’s Hypothesis/Model. The set K thus comprises diverse categories of dynamic types, such as $K = \{K_L, K_B, K_A, K_T, \dots\}$. This list is not exhaustive; many other nuanced coefficients may emerge from the interaction. However, these four provide an initial, formal vocabulary to analyze and intentionally design for a co-evolutionary dialogue that is the heart of retroductive play.

VI. DESIGNING FOR RETRODUCTIVE PLAY

Designing for retroductive play is an exercise in engineering for emergent order. The designer’s role shifts from generating specific outcomes to implementing the reasoning capacities of

the two agents, *GP* and *GA*, and the properties of the channels (Φ, Λ) that connect them. The overall system's precise final form cannot be dictated in advance; like any emergent system, it arises from the interplay of its agents and their environment. We cannot offer a blueprint for what retroductive play will look like, only a set of initial design provocations, our hypotheses, aimed at cultivating the conditions for its emergence.

The first provocation is to *target reasoning, not just performance*. The core operational mode of a retroductive game is that its adaptive system targets the player's reasoning process itself. The *GA*'s default Epistemic Target (*T*) should be the player's Hypothesis/Model. This reframes the purpose of adaptation away from simply managing success rates. When a player is stuck, the system's interventions should function as recovery scaffolds that aid cognition. Instead of just lowering difficulty with a u_C intervention, the *GA* might introduce a surprising fact – an unexpected piece of environmental feedback or a new tool – that is precisely engineered to challenge the player's flawed hypothesis and catalyze the abduction of a more insightful one. To effectively manage this, the *GA* must be engineered with a model of its policies and a capacity for meta-reasoning, allowing it to reflect on the success or failure of its adaptive strategies over time.

This epistemic goal depends on the protocol that governs the dialogue. Thus, a second provocation is to *engineer a legible and negotiable dialogue*. A robust dialogue requires a clear protocol, which depends on the system's channels and rules of engagement. The system requires high-bandwidth channels (K_B) to not only gather rich telemetry but also to make its reasoning legible to the player when desired. While high transparency can risk breaking immersion, this bandwidth is a capacity to be tuned, not a constant broadcast, even woven into the game's fiction through in-world cues, and character dialogue. Trust in the adaptive system must be earned, which means player opt-out and negotiation must be engineered as first-class features. The player must have the ability to reject or ignore interventions, ensuring the *GA*'s Authority is not absolute. Additionally, a dialogue is only as interesting as the vocabulary of its participants. The system must provide a rich action space for both the player and the *GA*.

A third provocation is to *value expressive inefficiency and embrace uncertainty*. A retroductive *GA* must possess a broad definition of meaningful play. Its perception module (Ω) should be designed to recognize "inefficient" but expressive actions such as creative stacking of boxes instead of solving a puzzle as valid player intentions, not errors to be corrected. If the *GA* detects persistent, intentional player action that conflicts with its hypothesis of an optimal path, it should be engineered to yield, treating the player's intent as a co-equal contribution. The degree of this yielding is genre-dependent; a sandbox game might unlock new creative tools in response, while a competitive game might flag the novel behavior for meta-level analysis. This also means the *GA* must leverage its uncertainty. Unexpected player action is not a bug but valuable data, prompting the *GA* itself to enter an abductive loop to revise its player model. A sophisticated *GA* could hold

multiple, contradictory hypotheses about the player's cognitive state and use targeted interventions as probes, effectively asking, "Are you this kind of thinker, or that kind?"

A fourth provocation then is to *treat the game as an evolving, malleable system*, where its very constitution becomes an object of the ongoing play dialogue. The system can be engineered to tap into the collective intelligence of the player base by integrating "play-the-culture" directly. By surfacing novel community strategies or highlighting solutions from diverse players, the system allows the meta-level retroductive loop of the entire culture to feed back into and enrich the individual's play-as-thought dialogue, closing the largest reasoning cycle.

Finally, *treat design as a retroductive process*, effective patterns will emerge from future player–system dialogues.

VII. CONCLUSION & FUTURE WORK

This paper frames gameplay through Peirce's concept of retroduction [1]. We explored how reasoning plays out in games—first via the player's loop, then by viewing the game as a retroductive agent. From this symmetry, we modeled player-game interaction as a reciprocal inference system and proposed design provocations inspired by this dual-agent perspective. The result is a vision of gameplay as a co-evolutionary dialogue between player and game—two retroductive reasoning agents. By viewing both as engaged in cycles of abduction, deduction, and induction, we offer a new lens on game design and AI that centers co-adaptation as a core design goal. This vision is both pragmatic and poetic: it provides a precise formalism to analyze, compare, and construct adaptive systems; while reimagining games as epistemic negotiations—where players and systems co-author possibility. The implication is not just better games that know the player, but that help players know more: about systems, about themselves, and what could be. A retroductive game does not merely adapt to the player, it dares the player to think differently next time.

This perspective invites new research directions. AI-for-Games could shift from winning to co-learning, rewarding agents for aligning their models with players' self-understanding. Human-computer interaction research might explore how to make a game's evolving mental model visible and interactive, inviting players to bluff, guide, or correct it. Procedural generation can shift from serving the player to collaborating with them, using level chunks to test hypotheses about player behavior, turning runtime design into a Bayesian experiment. And serious games could go beyond hints, delivering targeted counter-examples that provoke conceptual change.

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